		L	T	P	Cr.
B.Tech. (II Sem.)	**20CS05 - PYTHON PROGRAMMING**	**3**	**0**	**0**	**3**

Pre-requisite : Problem Solving Skills

Course Educational Objective: The objective of the course is to provide basic knowledge of python. python programming is intended for software engineers, system analysts, program managers and user support personnel who wish to learn the Python programming language for Problem solving and programming capability.

Course Outcomes: At the end of this course, the student will be able to

CO1: Identify basic building blocks of python to solve mathematical problems. **(Understand-L2)**

CO2: Apply the in-built data structures like list, tuple, set and dictionary for solving problems. **(Apply - L3)**

CO3: Use exception-handling mechanism to catch run-time errors **(Apply - L3)**

CO4: Demonstrate compelling concepts about string manipulation, regular expressions, and file handling. **(Understand - L2)**

CO5: Demonstrate object-oriented programming principles of python. **(Understand - L2)**

UNIT-I

Introduction to Python: History of Python, Usage of Python interpreter, Python Shell, Indentation, Python Built-in types, Variables, Assignment, Input-Output Statements, Simple Programs, Identifiers and keywords, Literals. **Operators:** Arithmetic operators, Relational operators, Logical operators, Assignment operators, Bit-wise operators, Python Membership Operators (in & not in), Python Identity Operators (is & is not), Operator precedence.

Control Structures: Conditional Statements - if, if-else, Nested if-else. Jumping Statements - continue, break, and pass. Python Loops - while, for, Nested loops with Programming Examples, Mathematical Functions and Constants (import math), Random Number Functions.

UNIT-II

Lists: Concept, Creating and Accessing Elements, Updating & Deleting Lists, basic List Operations, Reverse, Indexing, Slicing and Matrices, Built-in List Functions.

Tuples: Introduction, Creating & Deleting Tuples, Accessing values in a Tuple, Updating tuples, Delete Tuple Elements, basic Tuple Operations, Indexing, Slicing and Matrices, built- in tuple Functions.

Sets: Concept, Operations.

Dictionaries: Introduction, Accessing values in dictionaries, working with dictionaries, Properties, Functions.

UNIT-III

Functions: Defining a Function, Calling a Function, Types of Functions, Function Arguments, Anonymous functions, Global and Local Variables, Recursion

Modular Design: Creating modules, import statement, from, Date and Time Module.

Exception Handling: Exception, Exception Handling, except clause, Try, finally clause, User Defined Exceptions.

UNIT-IV

Python strings: Concept, Slicing, Escape characters, String Special Operations, String formatting Operator, Triple Quotes, Raw String, Unicode Strings, and Built-in String methods.

Regular Expression Operations: Using Special Characters, Regular Expression Methods, Named Groups in Python Regular Expressions, Regular Expression with glob Module.
Files: Introduction to files, file operations- Read, write, and search

UNIT-V
Object Oriented Programming OOP in Python: Classes, 'self-variable', Methods, Constructor Method, Inheritance, Overriding Methods, and Data hiding.

TEXTBOOKS:

1. Reema Thareja, "Python Programming Using Problem Solving Approach", Oxford Publications
2. Python for Everybody: Exploring Data In Python 3by Dr. Charles Russell Severance, Sue Blumenberg

Reference books:

1. Gowrishankar S and Veena A, "Introduction to Python Programming", CRC Press, Taylor, and Francis Group – A CHAPMAN & HALLBOOK.
2. R. Nageswara Rao, "Core python programming", Dreamtech, 2017.
3. Y. Daniel Liang, "Revel for Introduction to Python Programming and Data Structures", Pearson Publications.